Phase 3B | Prototype and component definition

PULSIPHER FARMS / MOAPA VALLEY CORN MAZE

Christmas Passenger Train First-Coach Development

PF-TRAIN-S3B-R0 | 17 October 2026

Issue purpose: compare documented components, calculate common engineering load cases, lay out the occupied mockup and define the first-coach prototype and pilot-track work. **Preliminary - not for fabrication, construction, procurement release or passenger operation.**

Owner basis: **0% longitudinal grade**; three regular coaches and one separate accessible coach; diesel propulsion beneath historical styling; enclosed conditioned cabins; hydraulic service preference. North Pole position remains deferred, with passenger disembarkation and accessible boarding at both stops. Ride experience and business projections are outside this technical issue.

What advances in Phase 3B	What evidence releases the next work
One common individual-car upper calculation case for frames, bogies, brakes and couplings.	Responsible engineer accepts design masses, load combinations, fatigue and stability criteria.
Documented section comparisons, supplier requirements and component fit checks.	Controlled OEM dimensions, actual ratings, interfaces and installation requirements.
Dimensioned occupied mockups, prototype assembly sequence and measurement records.	Recorded mockup results, reviewed detailed drawings and accepted first-article tests.
Technical supplier-return forms and a farm-crew pilot-track package.	Confirmed rail/wheel match, track design/tolerances, crew training and independent release.

The original Phase 3A issue and retained Stage 2 CAD remain available. This release adds development evidence; its drawings do not retroactively approve earlier space models. No manufacturer quotation, stock commitment or project-specific approval is implied.

01 | Recommended first-coach development sequence

Order	Decision or work	Deliverable / responsible party
1	Accept common mass/load cases; choose the running-gear integrator route.	Engineer load schedule and supplier response for two prototype bogies/four axles; full train eight/sixteen excludes locomotive.
2	Build the ground-level occupied mockup before freezing the interior.	Farm crew and vehicle/accessibility reviewer record seating, knee space, aisle, doorway and mobility-device trials.
3	Resolve actual bogie, brake, coupling, floor and body interfaces.	Controlled dimensioned OEM returns; coordinated vehicle engineer model. Do not machine from catalogue illustrations.
4	Complete chassis/attachments and body engineering around those interfaces.	Member/joint calculations, released shop drawings, material schedule, weld procedures, tolerances and inspection traveler.
5	Build and measure one chassis/coach before repetition.	Specialist frame and running-gear work; farm-crew work as assigned by released scope; actual mass and reaction measurements.
6	Execute pilot track, static checks, empty and restrained-ballast tests.	Track/vehicle engineers set methods and measurable acceptance limits. No passenger testing before release.

Normal accessible boarding is developed as an engineered bridge interface at both stops. The onboard lift remains an alternate requiring its own doorway, threshold, frame and emergency-rescue integration. These are technical development choices, not final component purchases.

02 | Common mass basis and revision control

Mass basis	Regular coach	Accessible coach	Train
Original tare allocation	7,040 kg	7,535 kg	14,500 kg locomotive/hotel plus four cars
Original nominal loaded	10,340 kg	9,235 kg	54,755 kg
Original stress loaded	11,825 kg	9,865 kg	59,840 kg
Phase 3B upper calculation case	13,200 kg each	11,000 kg	65,100 kg incl. 14,500 kg locomotive/hotel

The upper case rounds the former 65 t sensitivity distribution upward to whole individual-car values. It resolves Phase 3A's mismatch between 11.825 t regular frame calculations and approximately 13.171 t regular brake allocation. **It is not an increased target tare, a rated gross mass or authority to add passengers.** Actual loads and tare are still to be established. A supplier must state whether a bogie rating means total supported vehicle mass, carried sprung mass, axle load or another quantity.

Retain the 306 x 104 in clear cabin, 400 in frame study envelope, bogie centers x=45/261 in, and provisional 28 in floor height for interface studies. Datum x=0 is the front inside wall; y=0 is the coach center; z=0 is top of rail unless a drawing explicitly uses the ground-level mockup datum. No new final structural depth is fixed by retaining that floor height.

Source statuses: **VERIFIED** is a stated publication fact; **CALCULATED** is arithmetic from disclosed inputs; **STUDY** is an analyst choice; **HELD** needs engineering or supplier evidence. Dimensions and product families with an unresolved fit are not purchase selections.

03 | Running gear and candidate frame engineering

PF-TRAIN-S3B-R0 · 17 October 2026 · **PRELIMINARY - NOT FOR FABRICATION OR CONSTRUCTION**

Carry an engineered, braked, two-axle bogie under each end of every coach. Develop the coach frame around **two continuous steel girders**, with lightweight bodywork that is not credited as a structural truss. Obtain a complete bogie design responsibility, rather than buying individual wheels that merely fit 36-inch track. Carry 12-inch-deep girders into packaging review, with 10-inch members as the shallower comparison. Neither is selected for construction.

1. Evidence and purchasing route

All sources below were accessed 2026-10-17; source IDs and limitations are preserved in the companion JSON. No supplier was contacted. Published capability establishes a research lead, not an offer, stock, price or application approval.

Priority / route	Verified public evidence	What remains to be obtained
1 - complete running gear from Western Train	FAQ lists 36-inch trains, custom drive-truck regauging, custom parts, and air brakes on passenger coaches. BF04	Written willingness to supply complete **unpowered coach** bogies; actual load/speed/curve/CG envelope; hydraulic interface evaluation. Its generic passenger claims do not rate these cars.
1 - complete running gear from Severn Lamb	Identified **Jupiter** page publishes 914 mm gauge, 50 m minimum radius and enclosed-coach option. BF05	Standalone supply, exact 914.4 mm interface, mass and suspension ratings. Jupiter train specifications cannot be transferred to our coach.
2 - specialist custom engineering/manufacturing	Alan Keef's current About page names **The Steam Workshop** manufacturing partnership/successor; Keef now emphasizes consultancy and track work. Its experience range includes three-foot gauge. BF06	Current contracting/design entity, relevant coach design, delivery and US support. Older Keef project photographs do not establish current supply terms.
3 - engineer-led domestic integration	UTCRAS publicly lists wheelset design/engineering, custom axles, truck assembly, NDT and load testing. BF07	Confirm 914.4 mm work, new-bogie design responsibility and low-volume commercial fit; do not infer complete bogie design from assembly capability.
Wheelset alternative	GHH-BONATRANS describes customer-specific wheelsets across gauges and calculation/test development. BF08 BF09	Explicit small-diameter 914.4 mm proposal, minimum quantity, load spectrum, brake interface and maintenance support.

Rank: buy engineered complete bogies first; engineer-led specialist fabrication second; farm fabrication of wheelsets, axles, bearings or bogie frames is not the development baseline. Existing used equipment is viable only with identifiable drawings, material/history, inspection and requalification. Schaeffler offers application-designed axleboxes and sealed railway bearings, but no bearing part number is selected BF10.

2. Loads supplied to the bogie engineer

The shared upper cases are **13,200 kg regular / 11,000 kg accessible**, giving **65,100 kg** with three regular cars and the 14,500 kg locomotive/hotel allocation. These are engineering study cases, not gross ratings or permission to increase tare. Nominal/stress masses remain 10,340/11,825 kg regular and 9,235/9,865 kg accessible.

Coordinates remain inches: supports x45/261, body −3…309, frame −47…353. Static reactions are RA=Σm(261−x)/216; RB=Σm−RA; multiply kg-equivalent by 9.80665 for newtons. The upper fixed allocation is 8,415 kg regular and 8,670 kg accessible after subtracting stress payload. It is centred and uniformly distributed for this screen, including the unassigned upper-case reserve. The model assigns gross mass to the body; detailed design must separate sprung and unsprung mass.

Case	Front / rear bogie kg-equivalent	Larger bogie ÷ 2, reference only
Regular upper, actual seat-row centres; centred crew	6,600 / 6,600	3,300 kg
Regular upper, all 33 people at front-half centroid x76.5	8,294.69 / 4,905.31	4,147.34 kg
Regular upper, six people at front platform x−25; remaining 27 concentrated x153	7,316.94 / 5,883.06	3,658.47 kg
Same, six at outer edge x−47	7,405.56 / 5,794.44	3,702.78 kg
Accessible upper, actual 11 seat centres, two 295 kg occupied-device loads at bay centres, centred crew	5,157.15 / 5,842.85	2,921.42 kg
Accessible upper, people/devices at front-half centroid x76.5	6,325.21 / 4,674.79	3,162.60 kg
Accessible upper, six people x−47; remainder x153	6,305.56 / 4,694.44	3,152.78 kg

Mirror the boarding/crowding cases at the other end. Crowding positions are transient sensitivities, not permission for standing travel. Concentrating the remaining occupants at x153 also tests frame bending; it is not a prediction of boarding behaviour.

Do not specify axles by gross mass/4. For bogie wheelbase wb, axle forces are Rbogie/2 ± My/wb; My includes support offset, suspension reactions and braking torque. A separate uncoupled-vehicle sensitivity at 0.8 m/s² and assumed 1.5 m CG height transfers **294.41 kg-equivalent** forward on the regular car, raising its front-half case to **8,589.09 kg-equivalent**. Coupler-height moments and actual suspension can change this. No dynamic amplification or fatigue spectrum has been approved.

For 16 occupied seats on one side plus one centred crew member, the **10,880 kg** upper-fixed-allocation screen has CG offset 0.1733 m. At 7.5 mph/R30 m/hCG1.5 m, a rigid-body calculation gives 24.78% of total vertical force on the inner rail. An intentionally severe all-33-person one-side/front-half sensitivity reduces it to 11.51%. These use gauge as a support-spacing proxy, exclude wind, track twist, cant and suspension, and **do not establish individual wheel unloading or safe speed**. The centred upper-case reserve is not ballast credit: lighter and partly loaded cases can govern stability. Provide the actual mass/CG and suspension model before accepting 36-inch gauge for this wide body.

3. Frame arrangement and numerical member screen

Study girders run x−47…353 at **y±34 inches**, with vertical strong axes. Crossbolsters at x45/261 deliver loads to the bogie centre/sidebearer system; draft boxes must transfer longitudinal forces into both girders. Floor/seat crossmembers, wall posts, platform edges, stairs and lift mounts require identifiable load nodes. No roof/wall strength is credited. Seat-row centres are x41,73,…265; the close relationship of rows x41/265 to bolsters x45/261 requires designed common nodes. The accessible side opening x9…51 overlaps the front bogie region.

Published Atlas section properties BF01, pages 2-3, underpin this comparison. ASTM A500 Grade C 50 ksi is a candidate material basis supported by STI BF02; obtain material certificates and an engineer-selected toughness requirement. The nominal ¼-inch wall uses the table's **0.233-inch design wall**. E=200 GPa is

the ordinary steel elastic-modulus input BF03; citing building references does not adopt a building code for this vehicle.

Candidate HSS, inches	I / S, in⁴ / in³	Published lb/ft	Two 400-in girders	Worst static stress	Worst deflection
8×4×¼	42.5 / 10.6	19.02	575.2 kg	220.3 MPa	50.6 mm
10×4×¼	74.7 / 14.9	22.42	678.0 kg	156.7 MPa	28.8 mm
12×4×¼	119 / 19.8	25.82	780.8 kg	117.9 MPa	18.1 mm

The executable model integrates EI v″=M, sets zero displacement at both bolsters, and uses σ=M/S. Lateral load divides between girders by 0.5±y/68, assuming rigid crossmembers. Fourteen cases include actual seating, boarding, one-sided crowding and **500 kg equipment point sensitivities replacing equal uniform mass**, at x153 or −3. The equipment is an analyst load, not a selected product. Checks verify equilibrium, support displacements and the independent uniform-load moment expression; extrema use a 0.04-inch grid.

The platform/centre-crowd case governs moment at **38.258 kNm per girder**. One-sided front crowding governs deflection, at the front frame end. The upper fully seated regular car alone gives 120.4 MPa/18.1 mm for the 10-inch girder. A centred, symmetric crossbolster idealization at the 8,294.69 kg bogie case has **35.124 kNm** moment across the 68-inch girder spacing. This does not check pivot bearing or local HSS faces. The one-sided front crowd case sends 21.810/59.533 kN into the front bolster at the two girder lines: **32.578 kNm roll moment** must reach a designed bogie roll-reaction path. A centre pivot alone is not a complete support definition.

Increasing 10 to 12 inches adds about **103 kg** to the longitudinal pair and reduces its modeled deflection by 37%. Two hypothetical full-width 110-inch members of the 12-inch section add 214.7 kg; their total with the girders is 995.5 kg, leaving only about 504.5 kg of the original 1,500 kg regular underframe allocation for all other frame structure. This is a mass reservation, not a cut list.

At the provisional floor z28, a **0.75-inch panel-only sensitivity** places girder undersides at z19.25/17.25/15.25 inches. Insulation, finishes, tolerances, suspension travel, wheel/brake sweep and pivot height remain unprovided. The old three-inch frame graphic is superseded by a depth study. Preserve cabin area by placing deeper structure below seating where feasible; raise the floor or engineer a different structural arrangement if the bogie package requires it. Do not notch girders to make an unapproved bogie fit.

These stresses are **not allowable-stress checks**. Combined axial/biaxial/torsional loading, local wall yielding, buckling, weld fatigue, jacking, lifting, collision/draft impacts, cyclic twist, emergency loads and deflection limits remain engineering work. Shared brake/coupler load cases must enter the frame model at actual attachment heights.

4. Procurement specification - complete first-train lot

Item	Quantity / required supplier return
Complete coach bogies	**8**: six regular, two accessible; request common family if the engineer confirms all cases. Include suspension, axleboxes, brakes, retention and maintenance access.
Wheelsets / wheels / axlebox assemblies	**16 axles in 16 complete wheelsets / 32 wheels / 32 axlebox assemblies**; included in bogies, not extra quantities. Traceable materials, fits, NDT and assembly records.
Loads	All table cases and mirrors; asymmetric loading and brake pitch transfer; distinguish supported sprung load, bogie gross reaction, axle/wheel loads and fatigue spectrum. Supplier states rated envelope and limits.
Geometry	914.4 mm evaluation gauge; x45/261 supports; 28-inch floor target; supplier supplies new/worn wheel diameter/profile, back-to-back, axle journals, wheelbase, pivot/sidebearer heights and 3D travel envelope.
Track / brakes	Evaluate 30/50/75 m radii and 5/7.5 mph, reverse curves, turnouts, twist and 0% grade target. Wheel profile/rail section must be jointly signed off. Braked axle and parking/emergency interfaces coordinated to brake package.
Deliverables	Signed load/interface sheets, fatigue/strength and wheel-unloading evidence, inspection/maintenance limits, spares, drawings, cost/lead-time/warranty proposal. Availability and price **unverified**.

5. Procurement specification - first prototype release

Item	Quantity / hold point
Prototype running gear	**2 complete bogies / 4 wheelsets / 8 wheels / 8 axlebox assemblies** for one instrumented regular-coach chassis. It becomes part of the eight-bogie lot if accepted; remaining production is six bogies.
Frame development	One reviewed frame, measured mass/CG, certified materials and specialist welding/NDT; ballast fixtures and instrument locations designed before testing. No public passengers during development.
Bench qualification	Supplier-approved vertical/asymmetric load, suspension travel, retention, bearing rotation and brake tests. Engineer determines loads, cycles, limits and instrumentation.
Vehicle qualification	Controlled-ballast static deflection/corner-weight tests, curve/twist clearance, wheel-load assessment and braking tests on surveyed track; accessible lifting/boarding attachments require separate representative verification.
Production gate	Resolve measured strain/deflection against analysis, inspect joints, reconcile actual tare and wheel loads, accept wheel/rail interfaces, then authorize the remaining six bogies and three coach frames.

Farm crew can build full-size cabin/boarding mockups and prepare approved work areas. Structural welding, axle press fits, wheel machining, bogie alignment and safety-system qualification belong with qualified specialists under the responsible engineers' procedures. The next useful decision is a supplier-owned bogie interface drawing and accepted load-case matrix; it allows the floor/frame package to become buildable without sacrificing interior space.

04 | Brake and connection definition

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Decisions advanced

Use one consistent upper study consist: locomotive 14,500 kg, three regular coaches at 13,200 kg each, accessible coach 11,000 kg; total **65,100 kg**. These replace the previous inconsistent upper allocation; they are proposed engineering cases, not rated capacities. Nominal/stress budgets remain unchanged. **0% longitudinal grade governs.** Surveyed elevations and engineer-defined tolerances remain a hold; parking still addresses wind, inadvertent movement and residual profile deviations.

Retain distributed hydraulic braking as the preferred development path, with two isolated bogie zones per coach and locally available emergency/parking application. Advance **Knorr-Bremse HSC1P10N** as the smaller rail caliper for dimensional and dynamometer review. It is a spring actuator, not a pressure-to-apply service caliper. The supplier must either validate modulated spring braking for all required service duty or provide active service brakes plus an independently justified emergency/parking arrangement. Two release circuits alone do not establish independent safety paths.

Candidate	Documented facts	Decision
Knorr-Bremse HydroAct HSC1P10N	20 kN manufacturer brake force/disc; two hydraulic release circuits; maximum disc width 45 mm; approximately 15 kg [BCB01]	Advance for supplier engineering review. Force definition, mounting, release-pressure curve and repeated service duty held.
Hanning & Kahl HYS 150	12-18 kN stated clamp force at zero pressure; 35 mm disc width; disc diameter up to 350 mm; approximately 16 kg; −25 to +50°C [BCB02]	**No direct trailer-axle selection:** sheet specifies high-speed motor-shaft application. Obtain specific approval or an axle-rated alternative.
Hanning & Kahl HZY-K160-PR	160 × 168 × 350 mm; approximately 15 kg; 100 bar; proportional/stepped regulation and emergency function [BCB02]	Actual compact power-unit reference; not automatically compatible with another supplier's caliper.
Hanning & Kahl EHM	Independently activates two bogies; DC 24 V, control wire DC 48 V; 1100 × 510 × 500 mm; approximately 90 kg; 160 bar, IP65 [BCB03]	Useful complete-module packaging comparison. Check orientation and clearance; do not reserve an unverified installation beneath the 28 in floor.

No stock, price, lead time or configured compatibility is confirmed. Manufacturer force terminology must be resolved before comparing ratings. As an automatic-air alternative, DAKO documents a passenger-coach architecture using CV1nD automatic control, PZ4 emergency equipment, disc units and wheel-slide protection. [BCB04] Carry this alternative if a purchased locomotive and an integrator can provide a more supportable complete package. Air requires reservoirs, compressor/dryer, recharge/leakage evaluation and independent parking; hydraulic preference is not changed here.

Force, torque and thermal allocation

For preliminary allocation only, assume four braked axles per coach, with equal axle loads. Actual load distribution, mobility-device asymmetry, wheel unloading and braking layout supersede this division.

Vehicle	Service wheel/rail force, total / axle	Emergency force, total / axle	Emergency axle torque
Regular, 13.2 t	6.60 / 1.650 kN	10.56 / 2.640 kN	2640 r_w N·m
Accessible, 11.0 t	5.50 / 1.375 kN	8.80 / 2.200 kN	2200 r_w N·m

Here r_w is the **actual loaded rolling radius in metres**, not a selected dimension. Equal two-wheel allocation would halve axle contact force; actual wheel loads govern. Illustrative radii 0.20/0.25/0.30 m give regular emergency axle torques 528/660/792 N·m and accessible values 440/550/660 N·m. No wheel diameter is released.

For a disc, T = μ_pad N_sum r_effective, where N_sum sums normal force on both pad faces. A catalog's “clamping force” may use another convention. For an active brake, N_sum = P A_N, where effective area A_N includes piston area, lever/floating gain and efficiency; P_bar = 10 N_sum(N)/A_N(mm²). The 54-row parameter matrix varies wheel radius, effective disc radius and assumed pad friction, with illustrative effective areas. Regular emergency normal-force demand spans approximately 7.82-31.68 kN. Consequently, a nominal “20 kN” caliper is neither proven adequate nor rejected without its actual force definition and geometry.

A passive spring caliper requires its measured force-versus-release-pressure curve across wear, temperature and stroke; an ideal spring-minus-pressure equation is not a release setting. Pump flow must satisfy swept volume, hose/fluid compliance and leakage within the approved response time. Hydraulic pressure, line size, fluid, fittings and valve settings remain held.

The analyst targets b = 0.50 service / 0.80 emergency m/s² imply whole-train forces **32.55/52.08 kN**. At 7.5 mph with assumed 3 s equivalent delay, ideal level stops remain **21.30/17.08 m**. Required ideal all-wheel adhesion is 0.0510/0.0816; actual rail contamination and wheel-slide behavior remain unmeasured. Losing one complete regular coach's braking increases the emergency-command result to 18.87 m without redistributing torque; one regular bogie unavailable gives 17.88 m. These are fault studies, not permission to run with defects.

At 7.5 mph, translational stopping energy is **365.90 kJ** for the train; regular/accessible coach shares are 74.19/61.83 kJ, or 18.55/15.46 kJ per axle under equal allocation. Add rotational inertia. Gravity-related continuous heat is zero on the nominal level alignment; repeated stops and brake drag still require thermal proof. Temperature rise needs rotor mass, heat partition, material and cooling information.

Coupler loads and fit gates

Use the same launch resistance coefficient **0.010** across propulsion and connections, with acceleration 0.10 m/s² baseline and 0.15 alternative. These remain preliminary cases, not a complete load spectrum.

Joint	Trailing mass	Pull tension at 0.10 / 0.15 m/s²
Locomotive-regular 1	50.6 t	10.02 / 12.55 kN
Regular 1-2	37.4 t	7.41 / 9.28 kN
Regular 2-3	24.2 t	4.79 / 6.00 kN
Regular 3-accessible	11.0 t	2.18 / 2.73 kN

Reverse pushing produces the same quasi-static compression magnitudes for the same acceleration/resistance assumptions. For braking, C_j = ΣB_trailing − M_trailing ΣB_total/M_total, tension positive. The saved 32-row matrix covers balanced braking, each individual vehicle unbraked, one bogie unbraked and locomotive-only braking; reversing changes the sign for an otherwise identical allocation. Neither these results nor a blanket multiplier establish impact, fatigue or buff ratings. Coupler stroke, slack, transient brake buildup, curve compression and attachment eccentricity require the integrated engineer's analysis.

Prefer a semi-permanent railway connection between coaches. Dellner's People Mover assembly provides a real damper/pivot-anchor/cable-support reference, but its public page supplies no controlled installed dimensions or force rating. [BCB05] Voith 430/530 is a compact railway alternative advertised up to 300 kN tension/compression. [BCB06] Wabtec lists Compact Size I/II heads without public dimensions or ratings on the reviewed page. [BCB07] These are candidate families, not interchangeable parts. **No reviewed candidate presently passes the complete fit gate.**

The retained 58 in / 1473.2 mm pivot span, 30 in / 762 mm platform gap and 14 in / 355.6 mm setbacks are study geometry. To pass: obtain a configured anchor-to-anchor drawing and neutral/stroked lengths; verify yaw/pitch/roll and vertical freedom through all route and suspension cases; demonstrate platform/stair/guard and service-line clearance; identify maximum tension, compression, energy, fatigue and proof ratings; establish positive pin retention, wear limits and inspection access; and resolve the frame load path. Enlarge/revise the vehicle interface if necessary - do not cut down OEM drawgear. Pin sizes, welds and anchor plate thickness remain held.

Instrumented prototype and conditional acceptance

Build the first brake trial on a guarded dynamometer or secured rolling chassis, under specialist supervision. Use unoccupied cars and secured ballast, not passengers. Inspect and weigh the actual chassis before choosing ballast increments; proposed 25/50/100% load progression requires engineer acceptance of the gross load and restraint arrangement.

Instrument BTP01 demand timestamps; BTP02 local voltage/current; BTP03/04 front/rear actuation pressure; BTP05 auxiliary-release pressure/state; BTP06 actual actuator movement; BTP07 each axle speed plus independent ground speed; BTP08 disc/caliper temperature; BTP09 longitudinal acceleration; BTP10 emergency-continuity states. CTP01 measures connection force through a reviewed instrumented arrangement; CTP02 records drawgear stroke/pivot motion. Calibrate and synchronize channels, document uncertainty, and select sampling/anti-aliasing rates to resolve the shortest brake and coupling transient.

Trial	Measurements and conditional acceptance
Bench apply/release, new and worn friction parts	Measured torque must meet the approved axle demand over the supplier's accepted hot/cold and pressure envelope; residual drag and release time within specified limits.
Empty-to-ballasted service/emergency stops	Record initial speed, actual profile, delay, distance, deceleration, jerk and wheel slide. Distance plus measurement uncertainty must meet the **engineer-approved** limit; the ideal figures above are analysis references only.
One bogie/car response unavailable	Measure remaining forces and connection transients against the approved degraded case. No automatic permission for passenger operation follows.
Loss of power or ordinary data bus	Confirm locally sustained emergency/parking action, detected faults, traction removal and no automatic restart. Check actual force/movement, not only indicator lamps.

Trial	Measurements and conditional acceptance
Breakaway-equivalent input	First inject disconnect/open/short faults on a secured rig; then prove a detached restrained car applies and holds without locomotive energy. Moving separation requires a separately engineered containment plan.
Release-pressure trapped or valve failed	Compare actual actuator response to command; demonstrate the independently reviewed alternate stopping path or reject the architecture.
Parking and manual release	Apply a calibrated external load representing the approved parking case; measure movement for the approved duration. Demonstrate auxiliary-release isolation and restoration without uncommanded movement.
Repeated-stop thermal cycle and connection articulation	Establish pad/rotor limits, fade and recovery; sweep approved joint motions at controlled load without contact, hose tension, retention loss or unacceptable permanent set.

Numeric acceptance thresholds for timing, force, jerk, temperature, leakage, parking load/time and permissible joint movement must be approved **before testing**. Final valves, safety integrity, common-cause protection and software are not complete. Deliver supplier-controlled drawings, failure analysis, calibration records, raw traces, inspection results and signed release decisions before escalating from bench to closed-track trials.

Primary sources

Accessed 17 October 2026; product listings establish published specifications only.

- **BCB01:** Knorr-Bremse, HydroAct actuators, undated.

- **BCB02:** Hanning & Kahl, HYS 150 / HZY-K160-PR, September 2022.

- **BCB03:** Hanning & Kahl, EHM module, undated.

- **BCB04:** DAKO-CZ, Passenger transport brake systems, undated.

- **BCB05:** Dellner, People Mover semi-permanent coupler, undated.

- **BCB06:** Voith, Railway couplers, undated.

- **BCB07:** Wabtec, Compact Size I/II, undated.

05 | Interior mockup, body and accessible boarding

PF-TRAIN-S3B-R0 · 17 October 2026 · Development and mockup instructions; not production fabrication approval. Retain the Stage 2 cabin geometry and Phase 3B mass basis. Level track is now the owner's target; this does not establish platform height, cross-level, suspension travel or stair landing elevation. North Pole location remains deferred; equal boarding capability is required at both stops.

Build this mockup first

Start with the retained **32-seat, four-across regular coach**, followed by its 24-seat, three-across alternative. Use one taped floor and movable seating fixtures; there is no need to construct a complete coach to discover circulation problems. Test longitudinal seating third: it opens the center but changes passenger orientation and reduces the count.

Use a dry, level shop floor with approximately 40 × 20 ft clear work area. Mark a cabin rectangle 306 × 104 in, then the body outline 312 × 110 in and the full 400 in frame/platform envelope. The cabin diagonal is 323.190 in. Set the centerline and front datum first; measure both diagonals and record discrepancies. The target for **mockup layout only** is ±1/8 in at seat/door marks; this is not a production tolerance.

Coordinates below are inches: x increases from the front inside wall; y is transverse from centerline; mockup z=0 is the finished cabin floor. This differs from the vehicle drawings' rail datum: the provisional finished floor there is z=28 in. Seat top is z=18 in above cabin floor, matching the retained model; back/seat/foot geometry remains a trial envelope.

Element	Coordinate or dimension
Cabin	x=0…306; y=−52…52
Regular rows	x=31,63,95,127,159,191,223,255; each seat envelope extends 20 in forward in +x
Two benches per row	y=−52…−12 and +12…+52; each seats two in a 40 in allocation
Center aisle	y=−12…+12, throughout cabin
Regular end door	32 in clear, y=−16…+16 at x=0 and306
Platform reservation	x=−47…−3 and309…353; body outside y=−55…55
Six-adult starting positions	Four seated centers (106,−42),(106,−22),(106,22),(106,42); two standing/circulating centers (117,0),(181,0)

Tape all sixteen bench envelopes. Initially place real supported seating at x=95 and127 on both sides, so one occupied row faces another real seat back. Use stable commercially rated furniture suitable for the actual occupants, or fixtures designed and checked by a competent carpenter/engineer. Do not improvise seats from loose stacked boxes or treat cardboard as a handhold. Fixtures must match the envelope, including legs and backs; record any mismatch.

Materials/tools: two long tapes, chalk line, square/laser, contrasting floor tape, marker labels, camera, clipboard, four movable bench fixtures or equivalent chairs, lightweight braced door/side templates, actual coats and mobility devices. Templates can be cardboard or thin sheet on secured stands; they are visual boundaries, not supports. Mark the 86.614 in clear-height plane on a wall/laser; do not erect an unsupported overhead frame. Keep the first mockup entirely at ground level. Stair/lift trials use separately verified equipment later.

Space alternatives that fit the same cabin

Layout	Arithmetic and result	Main tradeoff to test
A: retained transverse	40+24+40=104 in; 8 rows ×4=32 places	20 in per adult; narrow aisle; 32−20=12 in between unoccupied envelopes
B: transverse with wide aisle	40+36+28=104 in; same rows ×3=24 places	Keep a two-person bench and generous single; aisle y=−12…24; existing32in end doors become narrower than aisle
C: seven roomy transverse rows	7 rows at34in pitch; x=41+34j, j=0…6; 4 per row=28 places	14in unoccupied gap; symmetric41in end zones
D: inward longitudinal	Two244in runs, x=31…275; 20in depth at each wall; floor center64in wide	At20in/adult,12 per side=24 places; two-sided9.5in knee trials leave45in between them; sideways riding requires separate safety review

The 9.5 in knee extension is a retained **geometry test**, not a human standard. Layout A leaves only 2.5 in between this trial extension and the next seat envelope. At the last row it leaves 21.5 in to the rear wall. An eight-row34in-pitch trial would shrink each end zone to24in and the rear occupied trial to14.5in, so it is not the preferred comfort alternative. Do not gain pitch by silently consuming egress space.

Trial A first with four seated adults and two circulating adults, then rotate participants. Record knee contact, foot placement, rising without pulling trim, shoulder/coats overlap, passing, end-door access and any forced sideways movement. Use a range of adult sizes; six people are a practical prototype group, not population validation. Standing positions illustrate circulation only. No standing-passenger capacity is added. Record measured minimum clearance, task completion, assistance needed and discomfort. Any blocked exit, loss of balance or inability to perform the task is a redesign trigger; numerical egress-time limits require the responsible designer’s criteria.

Accessible mockup and boarding

Retain the front60in turning circle centered at(30,6). The continuous36in route is x=60…306,y=−12…24; it is deliberately offset. Wheelchair bays are x=60…120 and120…180,y=−52…−16. Companion seats occupy x=80…100 and140…160,y=24…48. Rear rows are x=190,222,254: two-person benches y=−52…−12 and single seats y=24…52. Both42in side bridge openings are x=9…51. End openings are36in clear, y=−12…24.

Lead with a rated retained bridge at each station and train immobilization before passenger gates open. Platform height must follow the actual empty/loaded suspension range and approved gap treatment. Both stops must accommodate the same accessible coach; the North Pole’s final coordinates can follow later.

The optional BraunAbility NCL1000IB3754HB-2 remains **unintegrated**. Its published37×54in platform,1,000lb rating,45.85in minimum doorway and18in threshold depth are documented. A proposed48in clear opening at x=6…54 changes the door frame and needs structural and equipment approval. A conservative50.96in-wide,18in-deep inward reservation spans x=4.52…55.48,y=34…52; it overlaps the retained turning circle by2in at its extreme. This is a screening reservation, not the manufacturer’s exact stowed solid. Do not claim that widening the door alone solves fit. [BM05]

Use real power/manual chairs and a scooter with both bays occupied and companions seated. Include actual proposed belt/retractor footprints, emergency release and footrests. Check route, turning, shoulder alignment/adjacency of companions and both boarding sides; no nominal rectangle proves actual accessibility. Confirm governing rules before acceptance. [BM06, BM08]

Independent evacuation must remain possible with the onboard lift failed or its side blocked. Develop opposite-side bridge transfer to a separately rated mobile evacuation platform/lift on a firm reachable rescue corridor. Final survey must prove where that equipment can stand and reach every permitted train

stopping location. Test dead electrical power, manual lift operation and independent rescue with safe ballast before occupied demonstrations. No stair carrying of occupied wheelchairs is assumed.

Body material takeoff to develop with the fabricator

The following regular-coach takeoff uses retained model openings: sixteen22×38in windows, two32×78in end doors and86.614in wall height. Counts are material screens, not a released floor span or structural panel selection. All-aluminum honeycomb panels are only one body option; an engineered steel underframe/aluminum body remains the baseline.

Sample material role	Net coverage	Candidate typical mass	Calculated installed material
Cabin floor, outer footprint	238.33ft²	3/4in Plascore AA3.6-80,0.94lb/ft²	101.62kg
Outer wall panels after openings	380.10ft²	1/2in same panel,0.84lb/ft²	144.82kg
Inner wall liners after openings	365.66ft²	1/4in same panel,0.79lb/ft²	131.03kg
Flat roof panel proxy	238.33ft²	1/2in same panel	90.81kg
Inner ceiling proxy	221.00ft²	1/4in same panel	79.19kg
Interior finish	20.532m²	Tarabus Sirius NT,2.20kg/m²	45.17kg

Regular partial subtotal:592.65kg. Accessible partial subtotal:571.41kg, using its retained window and42in side-door openings; additional door framing/boarding equipment are excluded. It excludes framing, supports, insulation, glazing/doors, seats, external platforms/stairs, fasteners, inserts, adhesives, weather layers, curvature, overlaps and equipment. It is not the body or coach tare. Do not deduct apparent savings from the existing7,040kg regular tare allowance until the complete assembly is reconciled. The3in wall envelope leaves2.25in between the sample1/2in outer and1/4in inner panels before framing/insulation; thermal and fire performance are unproved. Roof curvature or a weather skin adds material. [BM01,BM03]

For a floor alternative,3/4in Coosa Bluewater26 over the outer footprint is175.67kg, adding74.05kg to this partial subtotal; equal thickness is not equal structural performance. Standard and fire-rated variants are distinct. Plascore edge closures and local hard points are essential. [BM01,BM02]

A zero-kerf geometric floor nest uses eight4×8ft sheets: six96×48in panels, three96×14in strips, two24×48in pieces and one24×14in piece. This demonstrates quantity only; actual supports, edge trimming, joints and grain/core orientation can change it. Do not cut/order from this nest. Finish can be screened as four cross-coach cuts2.6416m long from a2m-wide roll:21.133m² before seams/trimming versus20.532m² installed. Manufacturer seam/adhesive instructions govern. A single13mm insulation trial layer over outer walls/roof needs0.747m³; its mass awaits exact supplier data or weighing. No thicker assembly’s fire performance is inferred. [BM03,BM04]

First-article traveler and release order

Use one numbered traveler per sample module and first coach. Record drawing revision, serial/lot, material certificate, dimensions, actual mass, operator/date, inspector, deviations, photos and acceptance signature at every step.

1. Farm crew marks/records the mockup. Designer closes geometry decisions before structural cutting.

2. Supplier/engineer releases a representative floor-to-wall/window/roof joint module, with structural spans, inserts, joints and supports. Farm crew may make nonstructural cutting templates.

3. Specialist fabricates structure and hard points; inspect and photograph before coatings/closure. Record weld procedure and inspections; no invented weld sizes.

4. Dry-fit panels and actual door/window hardware; record diagonal, gap and minimum-clearance checks. Verify drainage and service access before bonding.

5. Qualify actual coated-substrate adhesive coupons, preparation, application temperature and cure; record batch/shelf life. Typical catalog adhesive strength is not a design allowable. [BM07]

6. Fit released insulation/liners/finish, keeping aluminum/steel isolation and intentional electrical bonding distinct. Farm crew can perform these trained processes; structural joints, critical equipment and tests remain specialist work.

7. Inspect sealed core edges, inserts, drainage and electrical access; conduct controlled leak/cleaning trials and record rework.

8. Weigh the module and finished car, reconcile every omitted item, then complete engineer-defined ballast, boarding-fault and rescue tests. Do not use people as proof-test loads or release passenger use from mockup success alone.

Sources were checked on **17 October 2026**; unchanged Phase 3A evidence is retained, and Plascore's table was rechecked for this takeoff: BM01 Plascore, BM02 Coosa, BM03 Gerflor, BM04 Armacell, BM05 BraunAbility application guide, BM06 US Access Board, BM07 Sika, BM08 Q'Straint. Product-specific limitations are recorded in the accompanying source register; no supplier acceptance, price, stock or lead time has been obtained.

06 | Locomotive and utility package interfaces

PF-TRAIN-S3B-R0 • 17 October 2026 • Preliminary engineering definition - not for fabrication or construction.

Locomotive selection route

Prepare comparable complete-locomotive offers against the actual coach masses below. Severn Lamb is the first technical benchmark because its catalog identifies a 914-mm vehicle; Western Train is the parallel domestic custom-build lead. Neither is selected or verified capable of this consist. Supplier acceptance must cover the four custom coaches, brakes and interfaces. Seat advertisements are not haul ratings.

Exact lead	Verified primary-source evidence	What the offer must resolve
Severn Lamb JUPITER	Catalog: 914 mm, 50 m minimum radius, 16 km/h, diesel/LPG, 1% maximum gradient [LU01]	Engine/calibration, working/adhesive mass, drawbar and wheel tractive-effort curves, thermal duty, brake integration and allowable custom trailing mass
Western Train Jupiter, custom 36-inch inquiry	Catalog Jupiter: 38-inch drive wheels, tender power, air brakes; photograph identifies a 24-inch version [LU02]. FAQ offers custom gauges including 36 inches and diesel/Eaton hydraulics [LU03]	Written identification and drawings of the proposed 36-inch variant; no hidden-coupling or horsepower inference from photographs
Alan Keef / The Steam Workshop	Current Keef page identifies consultancy/track focus and its manufacturing partnership with The Steam Workshop [LU04]	Which entity accepts locomotive design/build responsibility and US commissioning support; exact custom model/rating
Clayton custom diesel locomotive	Manufacturer publishes diesel range 4-150 t and multiple transmission arrangements [LU05]	Passenger-attraction controls, packaging, visibility, emissions, brakes and actual 914-mm configuration; industrial range is not a specific offer

Severn Lamb's Dayton custom electric locomotive demonstrates a separately engineered 3-ft platform, not diesel-Jupiter compatibility [LU06]. No public source located supplies the performance data needed to select a locomotive now. No supplier has been contacted, quoted or checked for available production slots in this phase.

Supplier performance return sheet

Project longitudinal grade is **0%**. As-built residual grades/settlement remain survey and engineering-tolerance holds. Proposed common upper load case: three 13,200-kg regular coaches + 11,000-kg accessible coach + 14,500-kg locomotive/hotel allocation = **65,100 kg**, including 50,600 kg trailing coaches. Nominal/stress totals remain 54,755/59,840 kg. These are engineering cases, not approved rated masses.

Upper case at 65.1 t	No acceleration	0.10 m/s² sensitivity	0.15 m/s² sensitivity
Starting wheel force, kN	6.38	12.89	16.15
Once-running wheel force, kN	3.19	9.70	12.96
Wheel power at 7.5 mph, kW	10.70	32.53	43.44
Engine screen, 75% drivetrain +10 kW auxiliaries, kW	24.27	53.37	67.92
Driven mass at assumed adhesion 0.15, t	4.34	8.77	10.98

Starting resistance=0.010×weight; running=0.005×weight; g=9.80665 m/s². Curve/wind resistance are **excluded and unquantified**. Acceleration values are alternatives to evaluate, not contractual acceleration minima or approved operating limits. At adhesion 0.10, the 0.15 m/s² starting case needs 16.47 t driven - greater than the entire locomotive/hotel allocation. Actual weight on powered wheels, fuel condition and contamination therefore matter even on level track. Hotel generation stays independent; the engine column does not include HVAC electricity. Equations and all three mass cases are saved in the portable calculator.

Require the supplier to return: continuous and short-time wheel/drawbar force versus forward/reverse speed; duration of every short-time rating; actual empty/full-fluid axle weights and powered fraction; cold/dirty bearing resistance; limiting radius and reverse-curve transitions; wheel/rail and coupling interface drawings; diesel emissions/application acceptance; radiator/oil-cooler performance at 110°F standstill; fuel consumption by duty; service access and exhaust limits. The 50-m Jupiter catalog radius is a candidate constraint, not a finalized site radius.

Brake return fields must cover the **whole consist**, wheel/brake balance, service/emergency/parking, loss-of-power/command, separation, residual-grade hold, traction inhibit and release sequence. Distinguish hydraulic **propulsion** from the owner's preferred hydraulic **brakes**; do not silently accept an air-only package as approval to change that preference. Couple braking and reversing loads into the draft-gear design.

Coach HVAC space claims: a measurable prototype decision

Retain regular/accessible thermal occupancy 33/14 and fresh-air screening flows 0.2475/0.105 m³/s, plus 0.03 m³/s door/leakage per car. At 110°F outdoors/75°F indoors the retained humid cooling screens are 22.70/13.28 kW. A supplier must return delivered sensible/latent capacity, ventilation and electrical input at those actual conditions; nominal refrigeration tons are insufficient.

Physical candidate	Published envelope and mass	Project fit decision
Konvekta HVAC 7403	3100×1600×415 mm; 520 kg; 30 kW cooling at 35°C outdoor/30°C indoor; 400VAC cooling/heating, 30-kW heater [LU07]	Regular-car roof **space claim only**. High-ambient capacity and 60Hz variant unverified. Whole 550-kg HVAC allocation leaves only 30 kg for mounts, ducts, drains and ventilation; mass fit is not established. Heater cannot be accepted at 30 kW/car without resizing the electrical study
Konvekta HVAC 5401	1500×1500×370 mm; 275 kg; 16 kW at 40°C/40°C; 400VAC, 7.2-kW heat [LU07]	Accessible-car comparison only; 450-kg allocation leaves 175 kg. Correct-condition cooling and cold-start heating still require verification
NW-BEOH-C + NW-64-CN split family	Evaporator 56-inch support width ×31-inch depth ×15-inch height [LU08]. Condenser image: 54-inch flange width ×38.437-inch maximum projection ×19.75-inch height [LU09-10]	Request custom 480V/60Hz package. Published condenser drawing is a single-phase example, not selected 480V hardware; weights and matched capacity missing

Study-centering the 7403 over x=153 inches gives x=91.98-214.02, y=±31.50 inches. With the existing **unengineered** 28-inch floor, 2.2-m clear cabin and 2-inch roof envelope, its top is **3.377 m above rail before curb/isolators**; the 5401 equivalent is 3.332 m. These are obstruction checks, not approved mounting locations. Reserve structural load transfer through roof rails/posts into the frame, independently of roof skin, glazing and decorative cladding; fastener pattern, reinforcement, fatigue, shock and lifting are

held for calculation.

Moving the same 520 kg from an assumed 1.0-m height to the roof-box midpoint raises a fixed 7040-kg tare car's calculated CG by 0.160 m; at 13,200 kg, by 0.085 m. This is **relocation sensitivity**, not actual vehicle CG. A centered 520-kg package contributes 2.55 kN to each bogie statically; this contribution is already within the mass allocation and must not be counted twice. Actual equipment CG and supports require supplier data.

The standard 15-inch evaporator below the current ceiling leaves only 71.61 inches headroom: reject that aisle location. A custom roof void/top-service arrangement or different unit is necessary. Underfloor condenser height leaves at most 8.25 inches above rail before structural/mount deductions; do not call it a fit until bogie travel, brakes, curves, suspension and clearance envelopes are checked.

Prototype utility interfaces and acceptance

Reserve distributed ceiling supply branches toward both cabin ends, separate return/filter access and a measured outdoor-air path. Duct depth/width must preserve standing clearance; final section comes from pressure/airflow/noise calculations. Show service-panel opening and component withdrawal volumes from supplier drawings before cutting body members. Route accessible condensate drains away from doors, steps, rails, wheels and electrical boxes, with trap/vent geometry from the selected unit. Avoid locating intakes in either diesel exhaust plume; no guessed separation distance is assigned.

Electrical prototype: protected temporary shore supply → approved transfer/distribution panel → local lockable disconnect → selected HVAC, lighting/chargers and essential battery supply. Final generator requires manufacturer-confirmed **continuous** output at actual ambient/altitude, kVA/power factor, motor-start/step-load voltage-frequency response, allowable loading, mass, cooling/exhaust and service drawings. Stadco 42/80 models publish 38/70 kW continuous under their reference conditions, not at 110°F [LU11]. Do not select from the model name. Final neutral/earthing, protections, cables and interlocked connectors remain qualified electrical design.

Before production release, approve measurable tests: weigh finished package and four wheel/axle reactions; inspect roof support load path and proof test only to engineer-issued criteria; traverse fresh/supply air with clean and specified dirty filters; measure cooling/heating, humidity and electrical input with simulated occupancy/door loads; record noise/vibration at seats and wheelchair locations; test drainage with approved motion/attitude cases and filter removal without disturbing seats. Record compressor start dips, shore movement inhibit, generator/shore interlock, bond continuity, insulation/protection and blackout emergency lighting/controls. Locomotive FAT uses approved ballast, calibrated force/speed/temperature instruments and both directions; compare with the supplier's guaranteed curves. Acceptance limits and fault-test methods must be agreed **before** tests.

Recommended next purchase-definition step: obtain manufacturer data for one regular-coach prototype HVAC package and complete locomotive offers; freeze voltage, equipment mass/CG and service envelopes before body detail release. All stock, lead times, ratings at site conditions and installation approvals remain unconfirmed. Source URLs and dated evidence are in locomotive_utilities.json.

07 | Pilot track and farm-crew installation

2026-10-17 · Preliminary engineering definition; not a construction or movement release.

Build crew competence around a small, inspectable track pilot before repeating work on the permanent railway. The proposed pilot uses the intended production rail, ties, fasteners and ballast after those items are engineered. It does not establish an allowable axle load, minimum route radius or operating speed. The project target remains **0% longitudinal grade**, verified by survey to engineer-issued limits. Drainage grades remain separate. North Pole placement does not block this pilot or vehicle engineering.

Reconcile the existing training guide

The current text of all 25 pages of Farm_Crew_Rail_Installation_Training_Guide_R0.pdf was read. Retain its supervised tool training, four-person task separation, rail handling, unloaded bend measurement, reference-rail setting, every-tie pilot checks, controlled surfacing, defect records and independent release gates. Preserve the original guide unchanged. Its 40-lb rail comparison and earlier 22-inch estimating tie spacing are **not selections**. Neither manufacturer bender capacity nor a classroom chord example establishes a permissible train curve.

This supplement adds defined planning modules, takeoff formulas, supplier questions and return records. It omits a physical transition because the modules are disconnected; a permanent tangent-to-curve transition still needs engineering. Illustrated pages were not visually reviewed in this reconciliation.

Pilot layout and measurements

Module	Planning geometry	Demonstration and records
A	60-ft nominal tangent, A0+00 to A0+60	Two nominal 30-ft lengths per rail; demonstrate one joint per rail near A0+30. Issued cutting, joint placement, gaps and joint-tie support govern.
B	24-ft reference arc, B0+00 to B0+24, study radius 50 m	Two separate bent rails. Verify unloaded curvature, springback and end tangency before fastening. Radius is a training choice, not route approval.

Keep these modules **physically disconnected** and isolated from operating equipment. They provide no braking runout, stopping protection or authority to roll a vehicle. An engineer must issue a separate restrained/load-test or movement plan before any such use.

Set protected survey controls and an offset line outside lifting areas. Record module coordinates and rail-head elevations against the same datum. At A, record control stations 0/10/20/30/40/50/60 ft; at B, 0/6/12/18/24 ft, plus curve ends and overlapping chord checks. Record gauge and crosslevel at **every tie** in this training pilot and at joints. Record twist using the engineer's specified base length; do not invent its limit. Check both rails' longitudinal profile, not only the centerline. Log gauge reference depth, instrument identification, calibration, rail temperature, load state and independent checker.

For the 24-ft, 50-m reference arc, $\theta = L/R = 0.146304$ rad $= 8.3826°$. A 10-ft chord has a 0.9146-in centerline mid-ordinate. These are geometry calculations, not tolerances. Rail measurement-line radii are $R \pm D/2$; D is the separation of the selected rail reference lines, **not automatically 36 inches**. Profile, cant and measurement plane affect it. Therefore inside/outside rail cut lengths differ; use the approved cutting schedule and bender end-crop allowance.

Material and tool decisions

Atlantic Track's railroad section sheet gives these nominal dimensions [TP01]. They identify different sections; they do not demonstrate their capacity for this train.

Section	Mass lb/yd	Height / base width	Head width	Web thickness
40 ASCE	40	3½ / 3½ in	1⅞ in	25/64 in
60 ASCE	60	4¼ / 4¼ in	2⅜ in	31/64 in

Request both as engineering alternatives with certified section drawings, steel properties, condition and matched joint/fastening systems. No rail is selected. The 65,100-kg proposed upper train case includes three 13,200-kg regular coaches, one 11,000-kg accessible coach and 14,500-kg locomotive allocation. Four-axle coach averages would be 3,300 and 2,750 kg/axle respectively; these are **not governing peak loads**. Rail selection needs actual wheel loads, imbalance, dynamic/lateral forces, contact geometry, tie support and subgrade response. Locomotive axle arrangement remains open.

The 84 route-ft pilot represents approximately 168 installed rail-ft. Six nominal 30-ft blanks provide 180 rail-ft before cuts: four for A and two for B. This is a planning stock schedule; supplier length availability, joint gaps, rejected ends and crop requirements can change it. Gross blank mass is 180 × section(lb/yd) ÷ 3: 2,400 lb for 40 ASCE or 3,600 lb for 60 ASCE. A single 30-ft blank weighs 400 or 600 lb; provide rated mechanical handling.

Count ties from the issued station schedule. A preliminary endpoint estimate is Σ[ceil(module length / maximum spacing)+1], amended for joints and actual end offsets. Plates equal twice the tie count only for a two-plate-per-tie system. Fasteners equal the issued pattern count per tie; joint bars and bolts follow the selected assembly. Ballast/subballast quantities come from issued section areas integrated along length, with separately stated compaction and procurement allowances. No spacing, ballast depth, spike pattern, bolt torque or thermal gap is fixed here.

Candidate	Documented capability	Purchase gate
Aldon #3H-50, item 4021-10	50-ton bender, listed 25-60-lb rail range; 137-lb tool [TP02]	Manufacturer confirms actual ASCE profile/grade, safe support method, current instructions and end crop. Tonnage does not establish achieved radius.
STANLEY SD67	Manufacturer manual: 7-10 gpm, 1,500-2,500 psi, maximum 250-psi return backpressure; 1,300-1,800 blows/min [TP03]	Confirm spike-head/tool-foot fit, exact variant, power unit, hoses and coupling compatibility; supervised training.
Aldon custom-size gauge inquiry	Manufacturer offers custom sizes; catalog 4022-02 is 56½ in [TP05]	Obtain a documented 36-in configuration, measurement plane, accuracy and calibration. No custom model yet verified.
Aldon #23-A	Documented 56-58-in range [TP06]	**Reject catalog version for this project**; “adjustable” does not mean 36-in compatible.

These are catalog checks on 2026-10-17, not stock or delivery confirmations. The September 2024 SD67 sheet identifies different weight/length variants [TP04]; specify the exact tool on the return form rather than transferring dimensions between versions.

Crew scope, hold points and procurement

Use the training guide's four task roles: lead/checker, machine operator, handler/tamper and spotter/second handler. Nobody measures beside moving rail. The farm crew can prepare access, receive/identify materials, construct released formation, place ties, assist supervised bending and fastening, ballast, tamp and collect records after training. The rail/civil engineer owns geometry, loads, material system and acceptance limits; a qualified surveyor establishes control; a track specialist demonstrates work and independently checks the pilot. Rail welding is excluded from this jointed pilot. Any later welded work needs a qualified specialist and approved process/inspection plan.

Hold	Evidence before release	Responsible reviewer
TP-H1 - before ordering	Signed wheel/rail interface; selected section, tie/fastener/joint drawings; tool and lifting methods	Rail/vehicle engineers; supplier application confirmation
TP-H2 - before covering formation	Survey, drainage path, geotechnical preparation and specified compaction results	Civil/track engineer and surveyor
TP-H3 - before fastening	Material identity; unloaded bend checks; joint/cut schedule; tool competence	Track specialist; engineer for deviations
TP-H4 - before surfacing acceptance	Complete fasteners, joints, gauge and line records; controlled lifting/tamping	Track specialist
TP-H5 - before repetitive crew work	Independent as-built checks, closed defects, demonstrated crew tasks	Responsible engineer/qualified inspector
TP-H6 - before loading or movement	Separate test plan, loads, restraint/runout, exclusion zone and post-load reinspection	Vehicle and track engineers; site safety lead

Procure in this order: obtain interface returns; release a single coordinated material system; reserve tools/training; buy pilot quantities only; inspect deliveries; build and close the pilot; then release repeat quantities against the final surveyed route. No supplier has been contacted.

The editable JSON forms include a material receipt, tie/station geometry record, bend-pass record, hold-point traveler, defect closure and two **UNSENT** technical supplier questionnaires. Each has blank result, evidence, reviewer and disposition fields. A blank limit or missing signature means **HOLD**, never pass.

The rail/wheel questionnaire requests profiles, back-to-back, contact/gauge references, loads, curves, wheelset tolerances, rail certificates, joints, fasteners, thermal design and installation instructions. The installer questionnaire asks who designs and signs, precisely what the farm crew performs, training and equipment, inspection hold points, deliverables, exclusions, availability and change control. It contains no price assumptions.

Both origin and North Pole still need coordinated boarding, accessible boarding and evacuation interfaces. Retain the Phase 3A coach-envelope calculation of 140 ft 10 in for four 400-in coaches and three 30-in gaps; it excludes locomotive and stop margins and does not prescribe full-length platform construction. Final civil layouts follow survey and vehicle-door positions.

Source URLs, dates, limitations, formulas and blank return forms are in track_prototype.json and track_pilot_field_forms_R0.json.

08 | Supplier and engineering return requirements

The accompanying editable forms are prepared for owner review and future authorized outreach. They have not been sent. A complete answer must distinguish standard hardware, project-specific engineering and items supplied by others.

Package	Required return before selection
Bogie/wheelset package	Exact model/options;914.4 mm gauge acceptance; new/worn wheel profiles and back-to-back; wheelbase, axle/bearing ratings, suspension curves/travel, bolster interface, brake mounts, mass/CG, minimum curve and reverse-curve limits, fatigue evidence and maintenance limits.
Brake package	Exact actuator/disc/pad/control set; force definition, pressure-to-torque curve, release/application pressure, temperature/friction limits, local stored energy, power/line-loss behavior, command/feedback interface, delay and validation evidence.
Rail coupling/drawbar	Operating and proof/ultimate force definitions; tension/compression/fatigue, installed height, articulation, slack/draft gear, retention, frame attachment, wear gauges and recovery provisions. Published force alone does not establish geometric fit.
Locomotive	Actual powered axle masses; drawbar effort at starting/continuous duty; speed/gearing, reverse duty, locomotive braking and train control, cooling/ambient/altitude limits, visibility, clearances, exhaust and service access for this consist.
Body/boarding/utilities	Controlled equipment dimensions, loads, mass/CG, mounts, duty ratings, clearances, fire/egress evidence, weather sealing, maintainability, power/ventilation/condensate interfaces and rescue requirements.
Track/system engineering	Matched rail/wheel/turnout drawings, axle loads, radius/transitions/crosslevel/twist limits, zero-grade survey acceptance, drainage/subgrade design, material criteria, permitted crew methods, inspection and commissioning plan.

Prototype quantities are planning counts only. Confirm the complete supplier boundary before procurement: a wheelset is not a bogie; a caliper is not a brake system; a quoted rail section is not an installed track package.

09 | Prototype records and release gates

Gate	Record to complete	Release authority / prerequisite
G1 · Occupied layout	Dated dimensions, representative adults/devices, passing/turning/door observations, rejected alternatives and revised CAD coordinates.	Vehicle/accessibility reviewer accepts requirements and changes. A successful mockup alone is not evacuation approval.
G2 · Components	Exact orderable options, controlled drawings, compatible ratings, supplied/excluded scope and source dates.	Vehicle/track/control engineers reconcile interfaces; owner authorizes purchase.
G3 · Fabrication	Material certificates, jig/datum and fit-up records, approved WPS/qualifications, weld inspection, dimensions, corrosion and maintenance access.	Released engineer drawings and specialist inspection plan; unresolved critical items block affected fabrication.
G4 · Static first article	Actual tare and axle/bogie reactions, dimensions/clearances, door/stair/boarding deployment, electrical/hydraulic prechecks and defects.	Responsible engineers accept proof method, instrumentation and limits; test ballast secured by an engineered method.
G5 · Dynamic/fault tests	Empty then restrained-ballast runs; speed/stop/pressure/temperature records; required failure cases, coupling articulation and no-power recovery.	Approved test track and procedure, exclusion zone and emergency plan; no passengers in engineering trials.
G6 · Repeated production	Accepted first-article configuration, closed defects, revised travelers, procurement list and inspection intervals.	Controlled release for repeated coaches; site-specific stations/alignment await survey and North Pole position.

The next revision should incorporate manufacturer responses and recorded mockup decisions, then close first-coach shop drawings. Final design responsibility, adopted standards, county classification, insurance conditions and acceptance criteria remain explicit project decisions. No outreach or orders were performed for this release.

10 | Primary source register

Accessed 17 October 2026 unless a record states otherwise. Manufacturer data support only the stated product and conditions. Exact project compatibility, price, stock and lead time remain unconfirmed unless explicitly recorded.

BF01 | Atlas Tube, ASTM A500 rectangular HSS section properties, pages 2-3

8/10/12 x4 x1/4 nominal dimensions, design wall0.233in, published nominal mass, A,Ix,Sx; PDF page3 visually inspected

Limit: Historical table in current manufacturer resource library; no current stock, mill tolerance, fracture toughness or vehicle design approval inferred.

https://www.atlastube.com/wp-content/uploads/2018/04/Atlas-HSS-Section-Properties-A500-Rectangles.pdf

BF02 | Steel Tube Institute, HSS Member Selection Tools: material and availability guide

A500 GradeC Fy50ksi, t_design0.93*t_nominal; production versus stocking distinction

Limit: Design aid for building/HSS selection, not vehicle load standard; exact supplier stock unverified.

https://steeltubeinstitute.org/hollow-structural-sections/hss-member-selection-tools/

BF03 | AISC ANSI/AISC358-22, Section13.5 elastic modulus definition

Official-source indexed excerpt specifies E29000ksi/200000MPa

Limit: Direct PDF fetch returned403; only indexed property excerpt verified. Used solely as material elastic constant, not as governing vehicle connection standard.

https://www.aisc.org/globalassets/aisc/publications/standards/a358-22w.pdf

BF04 | Western Train FAQ: gauge, custom trucks/parts and coach brakes

Explicit36in offering, customization including drive-truck regauging, custom parts, air brakes on passenger coaches

Limit: No published standalone unpowered coach-bogie rating, price, drawing, stock or fit for project; FAQ incorrectly calls gauge wheelbase in one answer.

https://www.westerntrain.com/faq

BF05 | Severn Lamb Jupiter, current identified product specification

914mm gauge; minimumradius50m; open/enclosed compartment options

Limit: Whole Jupiter product only. Gauge rounded914mm is not an accepted914.4mm tolerance. No transferred ratings or standalone-bogie availability.

https://www.severn-lamb.com/people-movers/tracked-trains/rail-trains/jupiter-2

BF06 | Alan Keef About: current manufacturing partnership and scope

Current site says track/consultancy focus and manufacturing partnership with The Steam Workshop as successor; experience10.25in-3ft6in

Limit: Broad experience includes3ft, not documented bogie rating or confirmed current contract capability. Older gallery evidence cannot establish present availability.

https://www.alankeef.co.uk/about-us

BF07 | UTCRAS Capabilities: truck assembly, wheelset design, custom axles and testing

Wheelset design/engineering, truckassembly, customaxles, NDT and loadtesting are listed

Limit: 914.4mm specific job, complete new-bogie design responsibility and small-order acceptance not published.

https://www.utcras.com/capabilities-1

BF08 | GHH-BONATRANS Design of new products

Customer-specific wheelset/component drawings, analytical/FE strength and test development

Limit: No project-specific dimensions/rating/quote; relevant standards must be selected by responsible engineers.

https://bonatrans.com/en/research-and-development/design-of-new-products/

BF09 | GHH-BONATRANS Wheelsets: customer-specific gauges and interfaces

Claims solutions across all gauges, tailored to bogie, brake, drive and track requirements

Limit: Marketing capability is not proof of an offered914.4mm small-diameter wheelset, minimum order or leadtime.

https://www.ghh-radsatz.com/en/products-and-services/wheelset/

BF10 | Schaeffler USA axlebox bearings for passenger cars and locomotives

Customer-designed railway housings; TAROL sealed greased assemblies; application-specific lubrication/testing

Limit: No selected bearing dimensions,C/C0,load spectrum or approved service interval.

https://www.schaeffler.us/us/products-and-solutions/industrial/industry_solutions/rail/axlebox_bearings_passenger_cars_locomotives/

BF11 | Valmont ASTM A500 tubing: square and rectangle properties

RectangularGradeC minimumyield50,000psi and tensile62,000psi corroborate BF02

Limit: Only cited rectangularmaterial rows; other site wording/olderround-property data not relied upon. Does not establish candidate section supply.

https://www.valmonttubing.com/resources-and-specs/astm-specifications/a500

BCB01 | HydroAct actuators

HSC1P10N: spring actuator; manufacturer force per disc20kN; two hydraulic release circuits; max disc width45mm; mass15kg.

Limit: Public documentation only; no quote, stock, lead time or configured project suitability verified

https://rail.knorr-bremse.com/en/gb/portfolio/products-and-systems/braking-systems/actuation/

BCB02 | HYS150 and HZY-K160-PR

HYS150 specified for high-speed motor shaft; passive clamp12-18kN at0bar, discwidth35mm,diameterup350mm,16kg,-25/+50C. HZY-K160-PR160x168x350mm,15kg,100bar; proportional/stepped regulation and emergency function.

Limit: Public documentation only; no quote, stock, lead time or configured project suitability verified

https://www.hanning-kahl.com/fileadmin/media/Downloads/ENG/Hydraulic_Brake_Systems/Federspeicher_HZY_150_EN_web.pdf

BCB03 | EHM electro-hydraulic module

Two-bogie independent hydraulic/electrical activation;24VDC with48Vtrainwire;1100x510x500mm;90kg;IP65;160bar.

Limit: Public documentation only; no quote, stock, lead time or configured project suitability verified

https://www.hanning-kahl.com/products/hydraulic-brake-systems/electro-hydraulic-module.html

BCB04 | Passenger transport brake systems

Passenger coach system names automaticCV1nD, emergencyPZ4, disc units, wheel-slide protection. Tram axle systems distinguish active trailer from passive motor-bogie brakes.

Limit: Public documentation only; no quote, stock, lead time or configured project suitability verified

https://dako-cz.eu/products/passenger-transport/

BCB05 | People Mover semi-permanent coupler

Double-acting damper,pivot anchor,cable support; configured dimensions/rating not published on inspected page.

Limit: Public documentation only; no quote, stock, lead time or configured project suitability verified

https://www.dellner.com/products/semi-permanent-couplers/semi-permanent-coupler-people-mover

BCB06 | Railway couplers

Type430/530 advertised compact rail coupling family, strengthup300kN tension/compression; configured dimensions unavailable on reviewed page.

Limit: Public documentation only; no quote, stock, lead time or configured project suitability verified

https://www.voith.com/corp-en/connection-components-couplings/railway-couplers.html

BCB07 | Compact Size I/II coupler head

Product listing confirmed; no controlled dimensions or force rating published on inspected page.

Limit: Public documentation only; no quote, stock, lead time or configured project suitability verified

https://www.wabteccorp.com/transit-rail/couplers-and-draft-gears/compact-size-iii-coupler-head

BM06 | US Access Board - ADA Chapter 10, sections 1002.2-1002.4

Amusement-ride rules specify approach, wheelchair space, companions, direct platform gaps and alternative bridge/ramp references. Side-entry advisory gives 42 in opening with 60 x 36 in maneuvering example.

Limit: Applicability to this train is unresolved; thresholds are not a blanket vehicle approval. Do not substitute advisory geometry for actual maneuvering verification.

https://www.access-board.gov/ada/chapter/ch10/

BM02 | Coosa - Bluewater Panels Imperial Technical Data Sheet

Bluewater26 nominal26lb/ft3; 4x8x3/4 in panel approximate52lb; fiberglass-reinforced polyurethane; standard panel FMVSS302 claim; material data typical, not application guarantee.

Limit: No approved train floor thickness/span, fastener strength allowable or complete fire assembly. Standard and fire-rated product variants must remain distinct.

https://coosacomposites.com/wp-content/uploads/2024/10/CoosaComposites_BluewaterPanels-Imperial.pdf

BM09 | Coosa - Flame Spread and Smoke Rated Panels

Separately ordered fire-rated Bluewater and Nautical lines; ASTM E84-23 test dated2024-09-23; published Bluewater flame spread10/smoke300 and ClassA claim.

Limit: E84 is not a rail or amusement assembly approval or proof of smoke toxicity compliance; obtain exact variant certificates.

https://coosacomposites.com/flame-spread-rated-panels/

BM01 | Plascore - Base Panels technical table

AA3.6-80 aluminum-skin/aluminum-core typical0.79lb/ft2 at0.25in,0.84lb/ft2 at0.50in and0.94lb/ft2 at0.75in. Open edges; manufacturer offers custom panels.

Limit: Typical test data; not structural allowables. Edge closure, inserts, skin support, adhesive/fire system and concentrated loads require project engineering.

https://plascore.com/composite-panels/base-panels/

BM04 | Armacell - ArmaFlex Rail SD technical specifications

Declared conductivity <=0.040W/(m K) at0C; vapor diffusion factor>=5000; published ASTM E162/E662 rail test claim limited to3-13mm; UV protection required; product-specific certified adhesives listed.

Limit: No weight/density adopted. Do not extend thin-product tests to thicker layered wall assembly; regional availability and complete US fire basis unconfirmed.

https://www.armacell.com/en-NL/product/technical-specs/armaflex-rail-sd

BM07 | Sika - Sikaflex-252 US Product Data Sheet, Version04.01 April2025

Vehicle assembly adhesive; actual-substrate adhesion/compatibility tests required; application10-35C; typical tensile lap shear2.5MPa; moisture cured.

Limit: Typical shear value is not allowable design stress. Surface prep, geometry, peel/fatigue/environment, cure and fire acceptance need qualified process.

https://industry.sika.com/dam/dms/us01/o/sikaflex-252.pdf

BM03 | Gerflor - Tarabus Sirius technical data sheet

NT2.25+/-0.15mm and2.20+/-0.15kg/m2; MK6.20+/-0.30mm and3.65+/-0.30kg/m2; R10 slip rating; FMVSS302 and ASTM E648Class1 statements.

Limit: Bus flooring candidate; assembly, substrate adhesive, wet farm dirt performance and applicable rail/fire requirements unapproved.

https://cdn.gerflor.com/media/2/38529/tarabus%20sirius%20-%20technical%20datasheet.pdf

BM05 | BraunAbility - NCL1000 Century Series Application Guide, cover 6-3-26 (3 June 2026)

Model NCL1000IB3754HB-2 row:385lb mass,1000lb rating,37x54in platform,66.06in overallheight,48intravel,50.96in overallwidth,45.85in minclear doorwaywidth,67in minclear height,18in thresholddepth. 12VDC/max70A; manual occupied raise/lower and fold/unfold backup.

Limit: Document cover6-3-26 is interpreted as3June2026 (US manufacturer); all dimensions reference only. Motor-vehicle lift certification does not approve rail mounting. Current42in side door conflicts.

https://www.braunability.com/content/dam/braunability/docs/comm/na/app-guide/updated-dpa-lift-app-guides/Lifts-Comm-Century-1000-Guide-6-3-26.pdf

BM08 | Q’Straint - QRT-360 wheelchair and occupant securement

Four-point retractable system; rail applications listed; manufacturer WC18/WC19 and30mph/20G test claims; Q-10007 kit includes4 L-track retractors and retractable lap/shoulder combination.

Limit: Must obtain complete anchorage/installation, actual wheelchair/rider mass limits, orientation and compatibility. Catalog test is not approval of custom coach anchor backing.

https://www.qstraint.com/qrt-360/

LU01 | Severn Lamb JUPITER

gauge_mm: 914; minimum_radius_m: 50; maximum_speed_kmh: 16; maximum_gradient_pct: 1; drive: ['diesel', 'LPG']

Limit: No exact engine, mass, adhesive load, TE curve or custom-coach rating. Catalog gradient is not project grade.

https://www.severn-lamb.com/people-movers/tracked-trains/rail-trains/jupiter-2

LU02 | Western Train Jupiter locomotive catalog

drive_wheels_in: 38; power_unit_location: tender; brakes: air; image_caption_gauge_in: 24

Limit: Not a verified 36-inch offer; photographs do not reveal coupling engineering.

https://www.westerntrain.com/locomotives

LU03 | Western Train FAQ

published_gauges_in: [15, 24, 36]; diesel_drive: diesel engine plus Eaton hydraulic pump; custom_regauging: offered

Limit: No exact model output/adhesive mass/grade or train-mass capability.

https://www.westerntrain.com/faq

LU04 | Alan Keef current capability and manufacturing partnership

focus: track and consultancy; manufacturing_partner: The Steam Workshop; narrow_gauge_range: 10.25 inches-3 feet 6 inches

Limit: Responsible manufacturing entity, exact custom model and US support must be confirmed.

https://www.alankeef.co.uk/about-us

LU05 | Clayton diesel locomotives and shunters

published_range_tonnes: [4, 150]; offering: custom diesel locomotives / multiple engines and transmissions

Limit: Industrial product range is not a specific passenger-attraction or 914-mm proposal.

https://www.claytonlocomotives.co.uk/diesel-locomotives-shunters/

LU06 | Severn Lamb Dayton custom electric project

gauge_ft: 3; motor_count: 4; motor_kW_each: 30; project_type: custom battery-electric

Limit: Not diesel Jupiter capability. Source battery power/energy wording ambiguous; not reused.

https://www.severn-lamb.com/dayton-celebration

LU07 | Konvekta rail HVAC overview - 7403 and 5401

HVAC7403: {'dimensions_mm': [3100, 1600, 415], 'mass_kg': 520, 'cooling_kW': 30, 'cooling_condition': 'ta=35C,ti=30C', 'heater_kW': 30, 'supply_VAC': 400}; HVAC5401: {'dimensions_mm': [1500, 1500, 370], 'mass_kg': 275, 'cooling_kW': 16, 'cooling_condition': 'ta=ti=40C', 'heater_kW': 7.2, 'supply_VAC': 400}

Limit: Catalog dimensions exclude fasteners; actual supports/CG not published here. Current production, refrigerant,60Hz and actual43.3C/23.9C performance unverified.

https://www.konvekta.de/fileadmin/user_upload/docs/datenblaetter/schiene/Produktuebersicht_Schiene_dt-engl_eBook_WEB_0821.pdf

LU08 | Northwest Rail Electric NW-BEOH-C evaporator

dimensions_in: {'support_width': 56, 'case_width': 52, 'airflow_depth': 31, 'height': 15}; top_service_option: available if ordered; matched_example: two NW-64-CN condensers

Limit: Mass and exact matched performance unverified;15-inch unit cannot be assumed to fit existing cabin ceiling.

https://nwrail.com/nwre_products/beoh-c.html

LU09 | Northwest Rail Electric NW-64-CN

nominal_cooling_tons: 5; normal_voltage: 480V three phase; variant_note: dimension illustration is single-phase240V; three-phase control box smaller

Limit: Nominal condenser rating is not whole-coach delivered cooling. Specific model suffix, dimensions and mass require supplier confirmation.

https://www.nwrail.com/nwre_products/64cn.html

LU10 | NW-64-CN manufacturer dimension illustration

support_flange_width_in: 54; height_in: 19.75; maximum_motor_projection_depth_in: 38.437; body_depth_in: 26.75; other_body_depth_with_parts_in: 30.827

Limit: Web extractor rejects GIF; directly retrieved official image and inspected locally. Single-phase example, dimensions vary per manufacturer; not a selected hardware installation drawing.

https://www.nwrail.com/nwre_products/64cn-dims.gif

LU11 | Stadco Mark6 sales specifications S32.25

6_42_continuous_kW: 38; 6_80_continuous_kW: 70; standard_3phase_V: [240, 480]; frequency_Hz: 60; rating_ambient_F: 80; rating_altitude_ft: 500; climatic_derating: supplier values required

Limit: Nameplate42/80 is not delivered continuous site rating. Price,leadtime and installation approval unconfirmed.

https://stadcogen.com/wp-content/uploads/2025/05/S32.25-Mark-6-Specs.pdf

TP01 | Atlantic Track: Railroad and Transit Rail Section Data Sheet

Primary supplier/manufacturer section table identifies 40 ASCE and 60 ASCE nominal weights and height, base, head and web dimensions. Table text read.

Limit: Dimensional identity only; no approved capacity, steel grade, stock or project suitability. PDF screenshot retrieval failed; unambiguous height/base/head/web entries used, ambiguous OCR head-depth entries excluded.

https://atlantictrack.com/wp-content/uploads/2020/08/Rail-Section-Data-Sheet-Railroad-and-Transit.pdf

TP02 | Aldon #3H-50 hydraulic rail bender, item 4021-10

Manufacturer lists 50-ton bender for 25-60-pound rail, 137-lb tool weight, supplied hand pump/hoses/fittings.

Limit: Confirm rail weight units, exact section and grade, instructions, springback/end-crop method and stock. Do not combine with older reseller catalog weight. No achieved-radius rating established.

https://www.aldonco.com/product/297-3h-50-hydraulic-rail-bender/

TP03 | STANLEY SD67 User Manual 17488, April 2024 version 15

Manufacturer manual hosted by distributor; p14 visually inspected: 7-10 gpm, 1500-2500 psi, 250-psi maximum backpressure, 1300-1800 blows/min, 1.9-in spike-head capacity. Operator instruction, hydraulic and training precautions read.

Limit: Specific variant, spike fit and complete power-unit/hose/coupler compatibility require supplier return and trainer acceptance; manual does not authorize construction use.

https://www.amronintl.com/mwdownloads/download/link/id/3027

TP04 | STANLEY SD67 specification sheet, September 2024

Manufacturer sheet visually inspected; identifies SD67101, SD67151, SD67111 and SD67161 variants with different length/weight. SD67101 short anti-vibration listed at 67 lb and 25.25 in length.

Limit: Do not transfer dimensions from another variant or conflicting retailer summary. Actual offered configuration and current availability unconfirmed.

https://www.amronintl.com/mwdownloads/download/link/id/3028

TP05 | Aldon aluminum track gauge, item 4022-02

Manufacturer catalog version is 56.5-in gauge, 6-lb nominal tool; manufacturer explicitly offers custom-size inquiries.

Limit: No 36-in model, accuracy, gauge measurement plane or calibration offer verified. Catalog item is incompatible as supplied.

https://www.aldonco.com/product/263-aluminum-track-gauge/

TP06 | Aldon #23-A adjustable level and gauge, item 4022-07

Manufacturer range 56-58 in; stated gauge accuracy 0.0625 in, nominal tool weight 7 lb.

Limit: Catalog version cannot measure 36-in gauge. Accuracy is not accepted as the project inspection requirement.

https://www.aldonco.com/product/266-23-a-adjustable-level-gauge/

TP07 | Farm Crew Rail Installation Training Guide R0 - owner reference

Current complete 25-page text read through Library; work methods, four crew roles and HP1-HP6 reconciled. Original preserved.

Limit: Owner teaching document, not an external product authority or approved construction specification; illustrations were not visually inspected in this reconciliation.